

Preliminary Exam Description
Programming and Quantum Computing
PGR-213 Spring 2026

Format: 15 Minutes Oral Exam

Topics: Quantum Error Correction and Quantum Communication

Procedure:

At the start of the exam you will “measure a qubit” in the Bell State $(|00\rangle + |11\rangle)/\sqrt{2}$ to determine your topic.

If the measurement result is $|0\rangle$ your topic will be “Quantum Error Correction”,
if the measurement result is $|1\rangle$ your topic will be “Quantum Communication”.

You will then have 15 minutes to present your topic using a whiteboard. Your presentation should include:

- a) A **concept map** giving an overview of the major concepts involved in your topic and how they relate to one another.
- b) An example **quantum circuit** illustrating an algorithm from your topic.
- c) An **example calculation** highlighting an important part of your circuit. Your calculation should demonstrate your ability to do basic calculations involving states and gates. Note that you will probably not have enough time to go through the entire circuit. It is recommended that you strategically choose a calculation from your circuit without going through the whole circuit.

The role of the examiners

The examiners may ask you questions throughout the exam, and can also help you out with hints if you get stuck.

Preparation Advice:

You can find lecture notes on quantum communication and error corrections on canvas. This will give you an idea of which concepts and algorithms are relevant for the exam. You can then use the notes and other supplementary sources of information to design your presentation. Fifteen minutes goes by very quickly, especially in an exam situation where you will naturally feel nervous. To help you deal with the nervousness it is recommended that you practice holding your exam presentation in front of your classmates. You can use a timer when you practice to get your talk to fit into 15 minutes.